

Final Project Report Template

1. Introduction

1.1. Project overviews

India's Agricultural Crop Production Analysis is a Tableau-based project that analyzes agricultural production data from Kaggle using an Excel dataset. It studies production by State, Crop, Year, and Season through interactive visualizations and dashboards.

1.2. Objectives

Analyze agricultural crop production.

- Compare production by State and Crop.
- Identify yearly and seasonal trends.
- Create an interactive Tableau dashboard.
- Present insights through a Tableau Story.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Raw agricultural data is difficult to analyze manually. Users need an interactive way to compare production across states, crops, years, and seasons.

2.2. Project Proposal (Proposed Solution)

Use Tableau to transform the Excel agricultural dataset into charts, KPI cards, maps, filters, an interactive dashboard, and a Story.

2.3. Initial Project Planning

Sprint 1: Data collection and preparation

- **Sprint 2:** Visualization development
- **Sprint 3:** Dashboard development
- **Sprint 4:** Story creation and testing

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Source: Kaggle Agricultural Crop Production Dataset

Format: Excel (.xlsx)

Fields used: State, Crop, Year, Season, Production.

3.2. Data Quality Report

The dataset was reviewed for missing values, duplicates, inconsistent data types, and inconsistent categorical values. Required corrections were considered before Tableau analysis.

3.3. Data Exploration and Preprocessing

The dataset was reviewed and organized for Tableau. State, Crop, Year, and Season were used for filtering and grouping, while Production was used as the main measure.

4. Data Visualization

4.1. Framing Business Questions

Which states have the highest agricultural production?

- Which crops have the highest production?
- How does production change by year?
- Which season has the highest production?
- How is production distributed across states geographically?
- What is the total production?
- What is the total cultivated area?
- How does production change using State, Crop, Year, and Season filters?

4.2. Developing Visualizations

State-wise Production

- Crop-wise Production
- Production Trend by Year
- Season-wise Production
- State Production Map
- Total Production KPI
- Total Cultivated Area KPI
- Average Yield by State
- Season-wise Pie Chart
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5. Dashboard

5.1. Dashboard Design File

Title: India's Agricultural Crop Production Analysis

The dashboard combines production charts, KPI cards, a map, and interactive filters for **State, Crop, Year, and Season**.

6. Report

6.1. Story Design File

Story Name: India's Agricultural Production Story

Story points:

1. Crop-wise Production
2. Production Trend by Year
3. Season-wise Production
4. State-wise / Average Yield Analysis
5. Final Interactive Dashboard

7. Performance Testing

7.1 Utilization of Data filters

Filters Tested: State, Crop, Year, Season

Result: Dashboard filters and interactions were tested successfully.

7.2 No of Calculation Field

Calculation Fields: Enter the actual number from your Tableau workbook.

7.3 No of Visualization

Visualizations: The project contains the charts, KPIs, map, and other visualizations listed in Section 4.2.

8. Conclusion/Observation

The project successfully converts agricultural production data into meaningful Tableau visualizations. It helps users compare states and crops, identify yearly and seasonal patterns, and explore the data interactively.

9. Future Scope

Add recent agricultural data.

- Add district-level analysis.
- Include weather and rainfall data.
- Add production forecasting.
- Add more detailed crop and yield analysis.

10. Appendix

10.1. Source Code(if any)

Not applicable – Tableau and Excel were used.

10.2. GitHub & Project Demo Link

github link - <https://github.com/akhareprarthana/India-Agricultural-Crop-Production-Analysis>